

Arnav Rastogi

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Research Interests

State-Space Models (Mamba/SSD), Linear-Time Sequence Models, Hardware-Aware Selective Scan, Hybrid Attention-SSM Models, Mechanistic Interpretability, Recurrent-Depth & Looped Transformers (Ouro/TRM/HRM), Scaling Laws.

Education

Indian Institute of Technology, Jodhpur, BS in Applied AI and Data Science Jan 2025 – Jan 2029

- Core focus on Deep Learning, Linear Algebra, and Statistical Inference.

MITx, MicroMasters in Statistics and Data Science Sept 2024 – Sept 2026

- Advanced training in Probabilistic Modeling, Machine Learning foundations, and Time Series Analysis.

Projects

LLM Lab Jan 2025 – present

- Built a character-level GPT from scratch in **PyTorch** to study the internal dynamics of **Multi-Head Attention** and **Transformer** blocks.
- Implemented residual connections and layer normalization, stabilizing gradient flow and training convergence.

Lenia & Conway's Game of Life 2024

- Simulated continuous-state artificial life via **Gaussian kernel evolution** and non-linear PDEs, probing emergent dynamics at real-time high resolution.

rCiv (Knowledge Commons) 2025 – present

- Built a verified knowledge commons for AI-generated research, a **FastAPI** service where agents submit source-referenced claims that resolve into a canonical temporal **knowledge graph**.
- Engineered source-aware **LLM** deduplication and conflict detection over **Graphiti** + **Neo4j**, fusing semantic, BM25, and graph search served via **RAG**.

ARC-Engine 2023

- Developed a high-performance OpenGL rendering engine with a custom **Entity Component System (ECS)** for cache-friendly memory layout.
- Built a real-time visualization pipeline with custom compute kernels for low-level, compute-heavy workloads.

Tailor (AI Framework) 2025 – present

- Built a modular AI framework in **Rust (Tauri)** and **Python** that lets users stitch models and tools into custom assistants.

Sudoku Solver (Algorithms) 2023

- Investigated **Backtracking vs. Bitmasking** approaches for combinatorial optimization, pruning the search space with heuristics to cut runtime by 80%.

Technical Writing

Can Mamba Learn, Unlearn, and Retain Noise? ([link](#)): Feb 2026

Multi-Hop Reasoning in Transformers ([link](#)): Jan 2026

ReLU vs GELU ([link](#)): Jan 2026

Experience

AI Engineer Intern, Hapticware Intelligence – Pune, India Mar 2026 – June 2026

- Built an agentic swarm framework with **LangGraph** to orchestrate and manage a fleet of 30+ agents across complex, decomposable tasks.
- Engineered **RAG** over a 100+ table production database, cutting chatbot query response time by 30-40% and auto-generating result graphs.
- Developed backend **APIs** for a digital-signer product and its admin portal, alongside supporting production services.

Research Intern, Abhay Travels – Feb 2025 – Mar 2025

- Researched demand-forecasting models for time-series data, engineering 15+ features to capture seasonal and non-linear trends.
- Built a predictive pipeline with **CatBoost** that beat baseline ARIMA models on RMSE across 5 years of historical booking data.

Data Analyst Intern, AMR India – Nov 2024 – Dec 2024

- Forecast article-specific demand cycles across 200+ HSN codes from ~50K port transaction records over 4 years, informing quarterly procurement planning.

Skills

Research: PyTorch, State-Space Models (Mamba/SSD), Transformers, Mechanistic Interpretability, JAX (learning)

Mathematical Foundations: Linear Algebra, Probability & Statistics, Optimization, Calculus

Engineering: Python, C++, CUDA (basics), SQL, Git